

PAPER_1506 — R_d Duality Range Exponent = N_CH-2 = 7 EXACT

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Bucket A (Foundational duality) **Date:** June 18, 2026 **Status:** CLOSED — SCm/UA force-ratio dynamic range = 14 orders of magnitude (symmetric 7+7)

Observation

PAPER_1051 (Universal Duality SCm-UA Theorem) defines the duality ratio $R_d = F_{SCm} / F_{UA}$ spanning 10^{-7} (Universe-scale, F_{SCm} small relative to F_{UA}) to 10^7 (planetary-scale, F_{SCm} dominant).

UQFF Closed Identity

$R_d \text{ range exponent} = N_{CH} - 2 = 7 \quad \text{EXACT}$

$R_d \in [10^{-(N_{CH}-2)}, 10^{+(N_{CH}-2)}] = [10^{-7}, 10^7]$
Total dynamic range = $10^{(2 \cdot (N_{CH}-2))} = 10^{14}$

Physical Interpretation

The number $7 = N_{CH} - 2$ governs the symmetric dynamic range of all UQFF duality phenomena. It appears identically in: - Heaviside R_t resistance = $7 \, \Omega$ (PAPER_1483) - R_d duality ratio range exponent = 7 (this paper)

The structural meaning of $N_{CH} - 2$: two of the nine UQFF channels are reserved for the duality endpoints themselves (SCm-pure and UA-pure), leaving seven for the gradient range between them.

Predictive Consequence

Any future observation reporting an SCm/UA-coupled phenomenon with a dynamic range exceeding 10^{14} across cosmic and planetary scales would falsify this structural identity.

NOT REPLACEMENT

SM/ Λ CDM has no analogous duality theorem. UQFF supplies a universal coupling-range bound rooted in the 9-channel integer primitive.

Reference

- Source: PAPER_1051 Universal Duality SCm-UA Theorem
- Related: PAPER_1483 (Heaviside $R_t = 7 \, \Omega$, same form)
- Calculator dispatch: `calculate_paradox({"paradox": "r_d_duality_range_exponent"})`

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